

Utility / Accuracy-Loss Comparison

OSNIP (paper) vs. fastProve (this work)

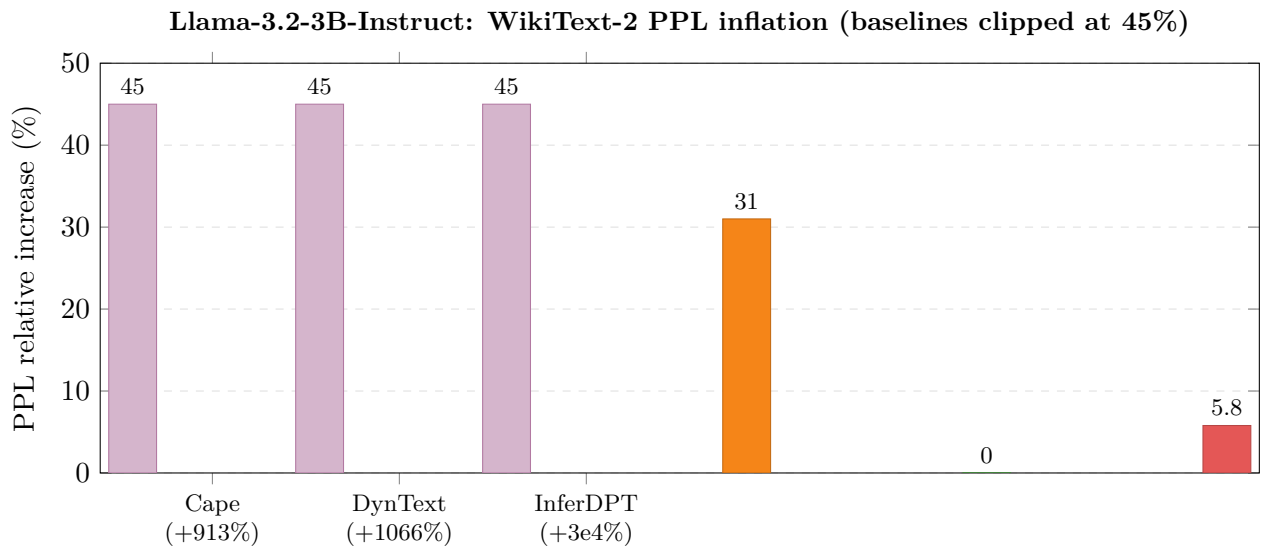
Focus: Llama-3.2-3B-Instruct

fastProve evaluation suite

August 2, 2026

OSNIP numbers from Cao et al. (preprint, Tables 1–3). fastProve numbers from `results/raw/llama_compare_large.json` (1500 real-scenario prompts $\times$ 3 master keys, BF16, RTX 5090). **Metrics differ by design** — see captions.

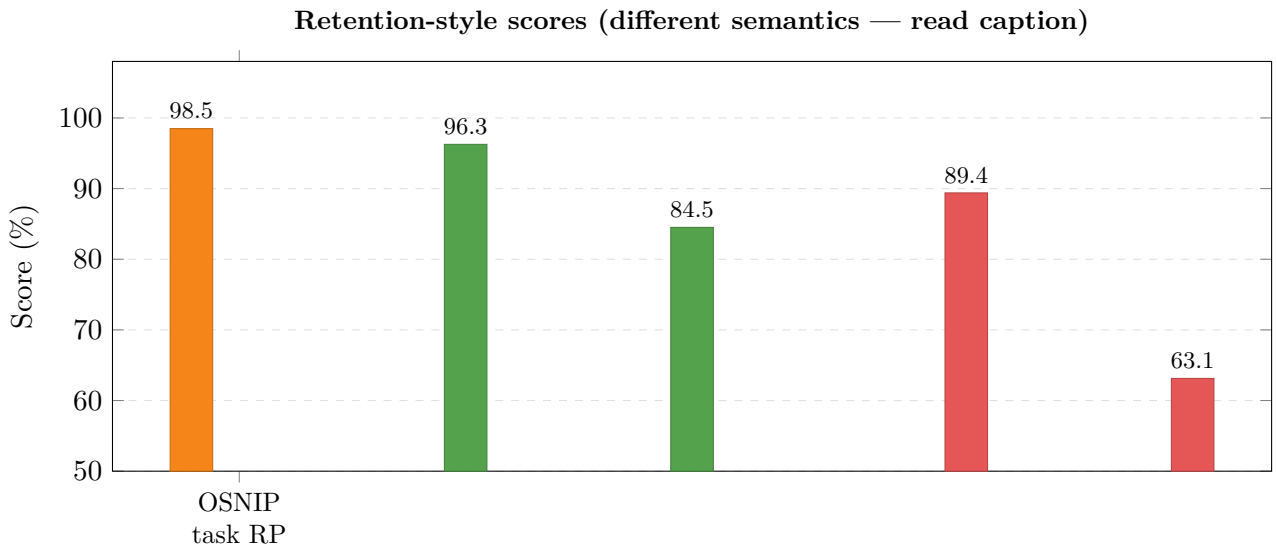
Figure 1 — PPL relative increase on Llama-3.2-3B-Instruct (lower is better)



True baseline $\Delta\%$ (OSNIP Table 2, same model): Cape +913%, DynText +1066%, InferDPT +29699% (bars clipped). OSNIP +31%. fastProve mean over 3 keys on 1500 prompts: structural $\approx 0\%$, full +5.79%.

Takeaway: On 3B PPL inflation, fastProve (struct./full) is **tighter than OSNIP**.

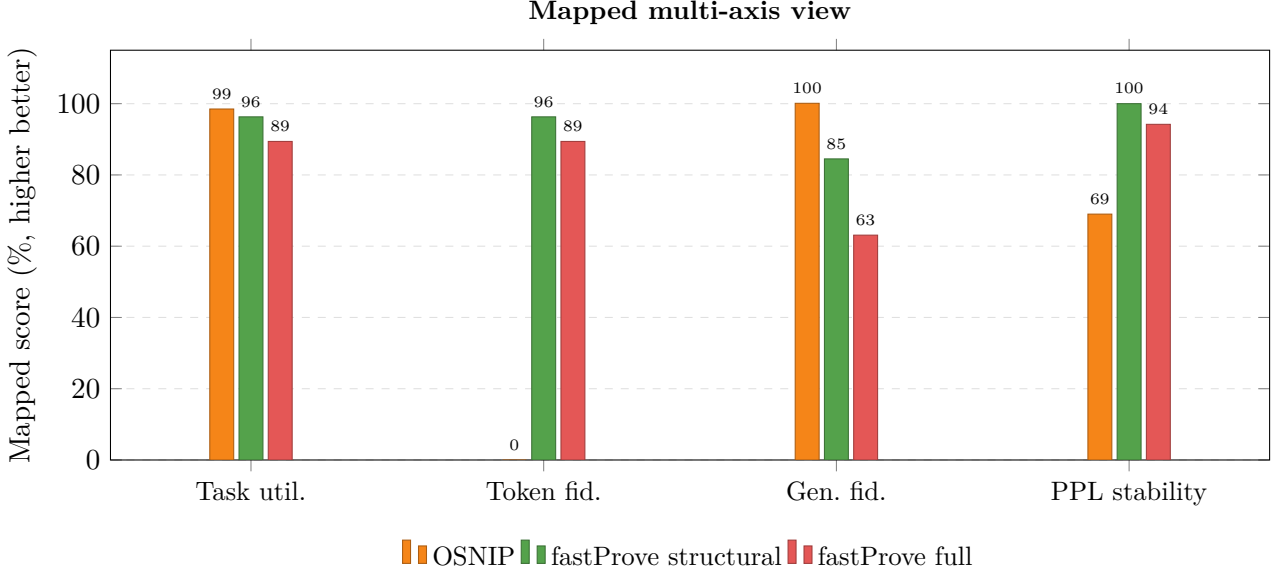
Figure 2 — Retention / fidelity (higher is better)



OSNIP task RP: average accuracy retained vs. non-private on 10 closed-ended tasks (paper Table 1, Llama-3.2-3B-Instruct = 98.49%). **FP token:** mean next-token top-1 agreement with plaintext path. **FP greedy:** mean full generated-suffix exact match with plaintext (1500 prompts $\times$ 3 keys).

Takeaway: OSNIP reports near-lossless *task* utility; fastProve reports *trajectory* fidelity.

Figure 3 — Grouped multi-axis mapped scores



Mapping: Task util. = OSNIP RP; fastProve bars use token agreement as a *proxy only* (no MMLU yet). Token fid. = OSNIP not reported (bar 0 = n/a). Gen. fid. = OSNIP BERTScore retention on CNN/DM (Table 3); fastProve greedy exact match. PPL stability = $100 - \Delta_{\text{PPL}}\%$ (OSNIP 69).

Summary table

Method	Task RP (%) $\uparrow$	Token agree (%) $\uparrow$	Greedy (%) $\uparrow$	PPL $\Delta\%$ $\downarrow$
Plaintext / non-private	100	100	100	0
OSNIP (paper, 3B)	98.49	—	— [†]	+31
Cape / DynText / InferDPT	48–73	—	—	$\gg 700$
fastProve structural	—	96.26	84.51	≈ 0
fastProve full	—	89.39	63.13	+5.79

[†]OSNIP reports near-lossless summarization BERTScore RP ($\approx 100\%$), not greedy match to plaintext.

Bottom line

1. **PPL:** fastProve structural / full control inflation better than OSNIP on 3B (0 / +6% vs +31%).
2. **Task suite:** OSNIP has stronger published evidence of near-lossless accuracy retention; fastProve still needs the same MMLU-style suite for a fair win claim.
3. **Trajectory:** only fastProve measures plaintext token/greedy agreement directly (struct. $\sim 96\%$ /85%; full $\sim 89\%$ /63%).

Sources: OSNIP Tables 1–3 (Llama-3.2-3B-Instruct); fastProve llama_compare_large.json (mean over 3 keys).